

CMDA Universal Analysis

<https://jpl-cmda.org/cmda.html>

User Manual

August, 2023

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

1. Select Service

Select Service:
Color Plot Number of datasets: 1

Click the box to see the pull-down menu of various services.

Search Filename/varNames/title: (info)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

Select Datasets:

Pick a basin:

Color Level Type (just an example)(info):

Specify Colormap:

Plot Title:

Analysis Purpose:

Select Datasets and Variables

Select Variables

Select Variables by Longname

There are 0 filtered datasets.

Number of selected datasets: 0

Display Options:

10 levels

rainbow

Modify It: 10

Analyse Data

Download Data

Browser URL:

Service URL

Browser URL

Open a page with the browserUrl

Data File URL:

Service Response Text:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

1. Select Service

Choose the service from the pull-down menu.

Search Filename/varNames/title: [\(info\)](#)
Filter by Variables Short Names
Filter by Variables Longname

Select Datasets:

Pick a basin
Color Level Type (just an example)([info](#))
Specify Colormap
Plot Title
Analysis Purpose

Analyse Data

Select Service:

- ✓ Color Plot
- Contour Plot
- Contour Plot Overlaying a Color Plot
- Subset and Download
- Download Original
- EOF: Empirical Orthogonal Functions
- Random Forest, Feature Importance
- Linear Combination of Multiple Variables
- Time Series Plots
- Map View of Multiple Variables
- Scatter Plot of Two Fields
- Difference of Two Fields
- Lagged Correlation of Two Fields

Number of datasets: 1

Modify It: 10

Download Data

Browser URL:

Service URL Browser URL Open a page with the browserUrl

Data File URL:

Service Response Text:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

2. Select Datasets and Variables

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

Select Variables

Select Variables by Longname

There are 0 filtered datasets.

Number of selected datasets: 0

Search datasets by typing keywords, variable short name or long name.

Select Datasets:

Pick a basin:

Display Options:

Color Level Type (just an example)([info](#)):

Specify Colormap:

Plot Title:

Analysis Details:

10 levels

rainbow

Modify It: 10

Analyse Data

Download Data

Browser URL:

Service URL

Browser URL

Open a page with the browserUrl

Data File URL:

Service Response Text:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

2. Select Datasets and Variables

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

nasa, airs

Select Variables

Select Variables by Longname

Filter Files

There are 3 filtered datasets.

Number of selected datasets: 0

Click "Filter Files".

Filtered datasets show up in a table.

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Pick a basin:

Display Options:

Color Level Type (just an example)[\(info\)](#):

10 levels

Modify It: 10

Specify Colormap:

rainbow

Plot Title:

Analysis Details:

Analyse Data

Download Data

Browser URL:

Service URL

Browser URL

Open a page with the browserUrl

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

2. Select Datasets and Variables

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

Select Datasets and Variables

nasa, airs

Select Variables

Select Variables by Longname

There are 3 filtered datasets.

Number of selected datasets: 1

Click the box next to the dataset you want to select.

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg):

form an axis

0

360

latitude (deg):

form an axis

-90

90

Time (year-month-day):

pick a value

2000-01-15

Preset Time Range:

At Beginning

+1 year

-1 year

At End

Dataset Time Range:

2002-09-15 to 2016-09-15

Pick a basin:

Data Subsetting Box shows up.

Display Options:

Color Level Type (just an example)[\(info\)](#):

10 levels

Modify It: 10

Specify Colormap:

rainbow

Plot Title:

Analysis Details:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

3. Subset Data

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

Select Variables

Select Variables by Longname

There are 3 filtered datasets.

Number of selected datasets: 1

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
1 ☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg):

form an axis ▾

0

360

latitude (deg):

form an axis ▾

-90

90

Time (year-month-day):

pick a value ▾

2000-01-15

Preset Time Range:

At Beginning

+1 year

-1 year

At End

Dataset Time Range:

2002-09-15 to 2016-09-15

Pick a basin:

▾

Choose data subsetting options.

Display Options:

Color Level Type (just an example)[\(info\)](#):

10 levels ▾

Modify It: 10

Specify Colormap:

rainbow ▾

Plot Title:

Analysis Details:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

3. Subset Data

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

Select Variables

Select Variables by Longname

There are 3 filtered datasets.

Number of selected datasets: 1

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
1 ☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg):

latitude (deg):

Time (year-month-day):

Preset Time Range:

Dataset Time Range:

Pick a basin:

form an axis

form an axis

✓ pick a value

average over

sum over

form an axis

0 360

90 90

2000-01-15

+1 year

6-09-15

-1 year

At End

Click the pull-down bar to choose action:

- pick a value
- average over
- sum over
- form an axis

Display Options:

Color Level Type (just an example)[\(info\)](#):

Specify Colormap:

Plot Title:

Analysis Details:

10 levels

rainbow

Modify It: 10

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

3. Subset Data

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: (info)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

Select Variables

Select Variables by Longname

There are 3 filtered datasets.

Number of selected datasets: 1

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
1 ☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg): form an axis

latitude (deg): form an axis

Time (year-month-day): average over

Preset Time Range:

Time Procession: ☐ Use Anomaly ☐ Yearly Mean ☐ Quarterly Mean

Type the values to determine the range.

Dataset Time Range: 2002-09-15 to 2016-09-15

Pick a basin:

Display Options:

Color Level Type (just an example)(info): 10 levels Modify It: 10

Specify Colormap: rainbow

Plot Title:

Analysis Details:

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

3. Select Display Options

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

Select Variables

Select Variables by Longname

There are 3 filtered datasets.

Number of selected datasets: 1

Select Datasets:

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
☐	data/data_clean/cmip5/nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc:	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
☐	data/data_clean/cmip5/nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc:	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	mouse hover	20020916	20110516	
1 ☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg):

form an axis

0

360

latitude (deg):

form an axis

-90

90

Time (year-month-day):

average over

2002-09-15

2016-09-15

Preset Time Range:

One Year

Add 1 Year

Reduce 1 Year

Full Range

Time Procession:

☐ Use Anomaly

☐ Yearly Mean

☐ Quarterly Mean

Dataset Time Range:

2002-09-15 to 2016-09-15

Pick a basin:

Display Options:

Color Level Type (just an example)[\(info\)](#):

10 levels

Modify It: 10

Specify Colormap:

rainbow

Plot Title:

Analysis Details:

Choose from pull-down menu and type values.

Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

4. Submit an Analysis Run

Select Service:

Color Plot

Number of datasets: 1 . Number of dimensions: 2

Select Datasets and Variables

Search Filename/varNames/title: [\(info\)](#)

Filter by Variables Short Names:

Filter by Variables Longname:

Filter Files

nasa, airs

tas ✕

Select Variables by Longname

There are 1 filtered datasets.

Number of selected datasets: 1

Select Datasets:

		fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list if number of vars > 1
1	☒	data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc:	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	

Data Subsetting

longitude (deg):

form an axis

0

360

latitude (deg):

form an axis

-90

90

Time (year-month-day):

average over

2002-09-15

2016-09-15

Preset Time Range:

One Year

Add 1 Year

Reduce 1 Year

Full Range

Time Procession:

☐ Use Anomaly

☐ Yearly Mean

☐ Quarterly Mean

Dataset Time Range:

2002-09-15 to 2016-09-15

Pick a basin:

Display Options:

Color Level Type (just an example)[\(info\)](#):

Specify Colormap:

Plot Title:

10 levels

rainbow

Modify It: 10

Click "Analyse Data" box.

Analysis Details:

Analyse Data

Download Data

Browser URL:

Service URL

Browser URL

Open a page with the browserUrl

Analyse Data

Download Data

Browser URL:

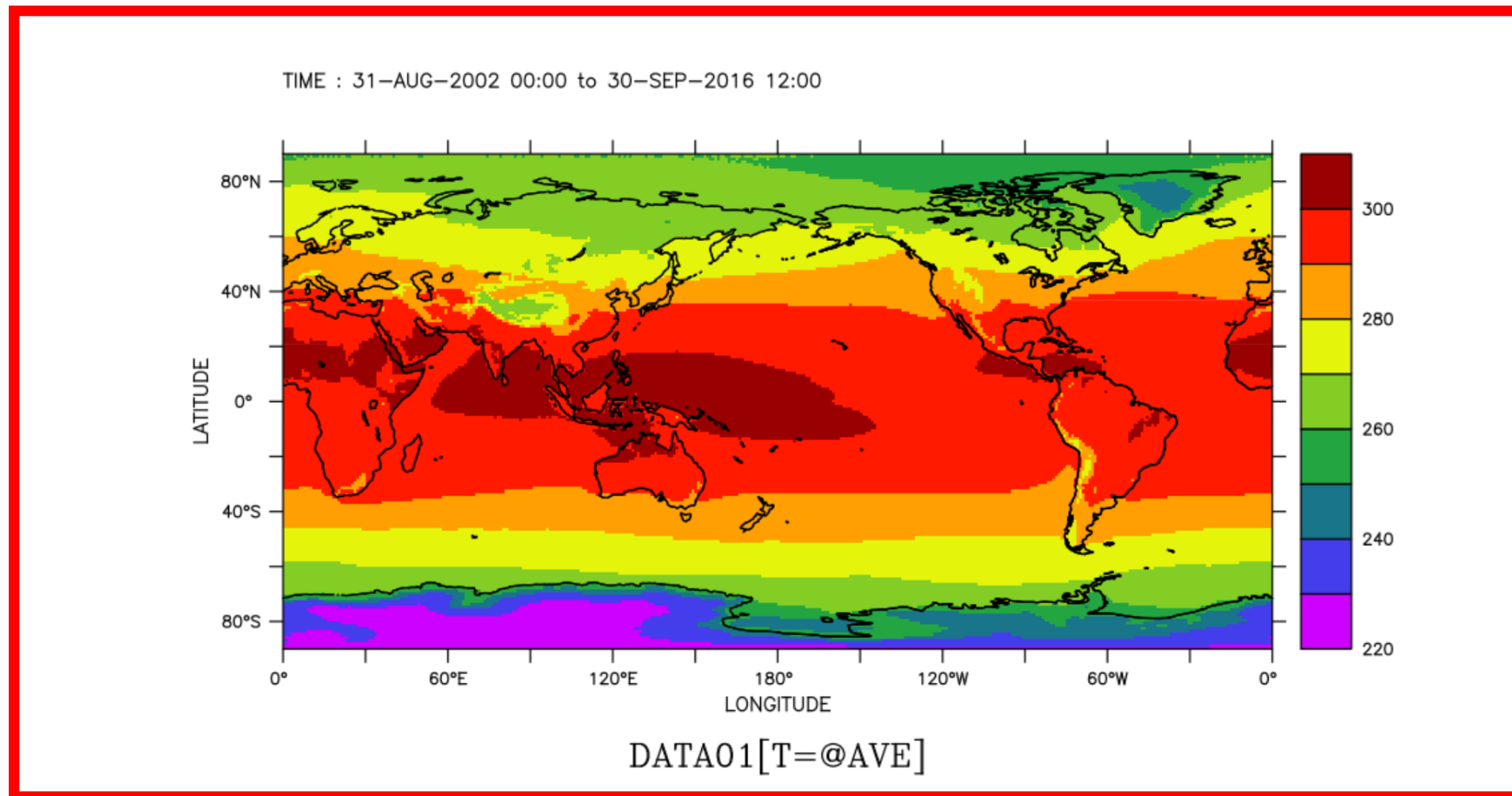
Service URL

Browser URL

Open a page with the browserUrl

4. Submit an Analysis Run

The analysis result shows up in the box below.
If the result is an image, you can save the image to your local server.



Analyse Data

Download Data

Browser URL:

Service URL

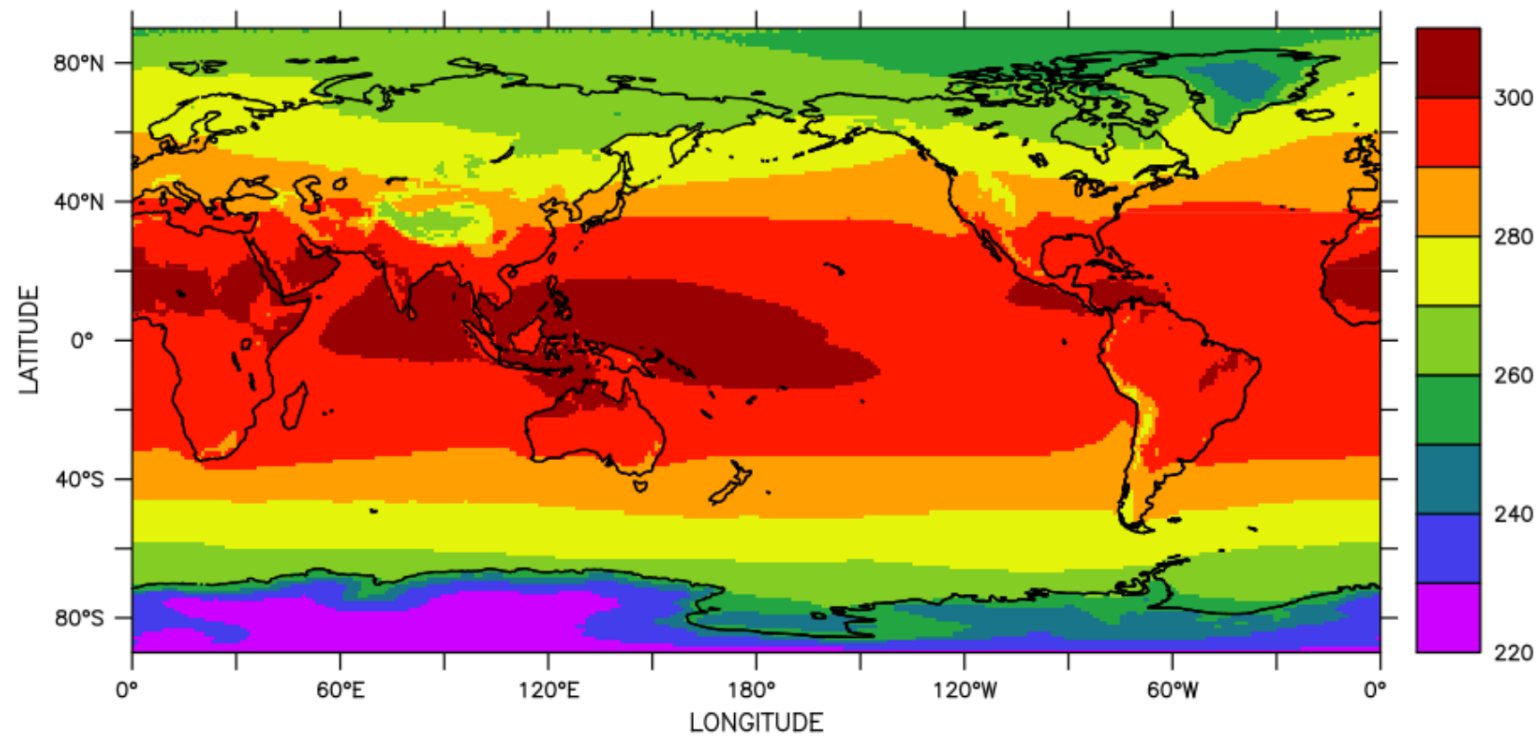
Browser URL

Open a page with the browserUrl

5. Download the Analysis Result

Click “Download Data” to download the analysis result in NetCDF file.

TIME : 31-AUG-2002 00:00 to 30-SEP-2016 12:00



DATA01[T=@AVE]

Analyse Data

Download Data

Browser URL:

Service URL

Browser URL

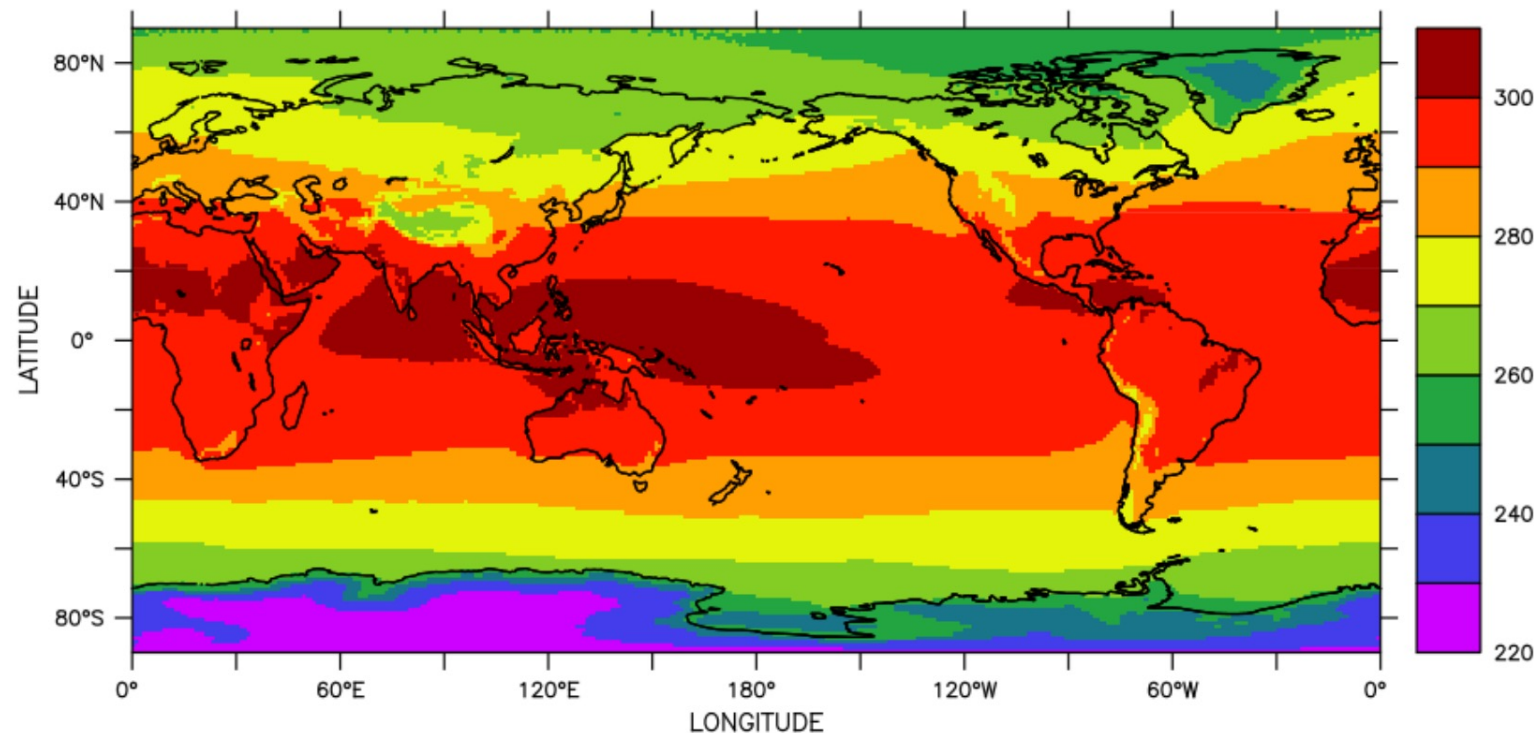
Open a page with the browserUrl

https://jpl-cmda.org/svc/universalPlotting6b?vars=tas&datasets=/mnt/data/data_clean/cmip5/nasa/airs/tas_AIRS_200209-201609.nc&lonMethod=3&lonS=0&lonE=360&latMethod=3&latS=-90&latE=90&presMethod=0&presS=500&presE=800&timeMethod=2&timeS=2002-09-15&timeE=2016-09-15&ferretLevel=10&colorMap=rainbow&plotTitle=&anomaly=0&climatology=undefined&timeMean=&purpose=&service=plot&basin=&userId=&serviceId=&dataUrl=https%3A//jpl-cmda.org/static/universalPlotting6b/554bcb828d85139b83594dc57724d220/data.nc&plotUrl=https%3A//jpl-cmda.org/static/universalPlotting6b/554bcb828d85139b83594dc57724d220/plot.png

6. Generate a Service URL text

Click “Service URL” to generate a service URL text, which you can copy over to an API call function in Jupyter Notebook.

TIME : 31-AUG-2002 00:00 to 30-SEP-2016 12:00



DATA01[T=@AVE]